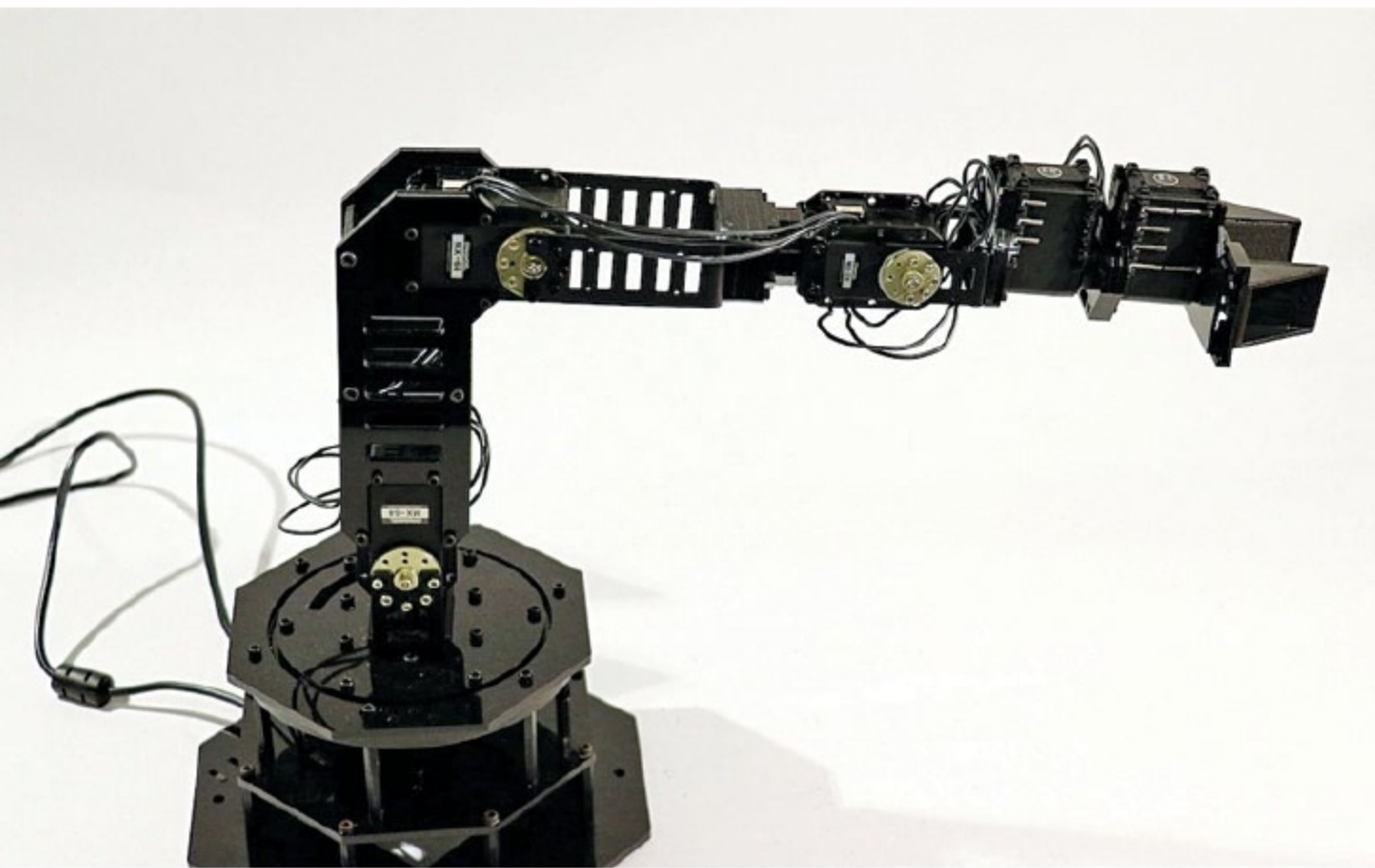


A robot that can imagine itself

Robots that are self-aware have been science fiction fodder for decades. But now we may finally be getting closer. Humans are unique in being able to imagine themselves—to picture themselves in future scenarios, such as walking along the beach

on a warm sunny day. They can also learn by revisiting past experiences and reflecting on what went right or wrong. While humans and animals acquire and adapt their self-image over their lifetime, most robots still learn using human-provided simulators and models, or by labourious, time-consuming trial and error. Robots have not learned to simulate themselves the way humans do.

Columbia Engineering researchers have made a major advance in robotics by creating a robot that learns what it is, from scratch, with no prior knowledge of physics, geometry or motor dynamics. Initially, the robot does not know if it is a spider, a snake or an arm—it has no clue what its shape is. After a brief period of babbling, and within about a day of intensive computing, it creates a self-simulation. The robot can then use that self-simulator internally to contemplate and adapt to different situations, handling new tasks as well as detecting and repairing damage in its own body.



Robotic arm that is used to perform tasks (Credit: Columbia Engineering)

First ionically-cooled wireless charger demonstrated at CES 2019

Integrated Device Technology (IDT) demonstrated at CES 2019, a cutting-edge wireless power transmitter that enables mobile OEMs and peripheral manufacturers to design high-capacity wireless charging into their products. Developed around IDT P9242-G converged Tx IC and coupled with Ventiva's line of silent, small, vibration-less air movers, the new solution shows how innovative and thoughtful product design can mitigate and even eliminate system-level heat and noise constraints to wireless charging once thought intractable. The demonstration is the product of collaboration and investment by IDT in Ventiva, a Silicon Valley-based startup developing solid-state air moving technologies.

According to IDT, the pre-production Tx can deliver a sustained 15W of input power to a 15W-enabled wireless power device while keeping the smartphone cool to the touch. In high-capacity mode, Tx charges a receiver device as fast as if it were plugged into an electrical outlet. Tx is compatible with existing wireless power receivers, and is also smart, capable of decreasing input power or increasing cooling as the device being charged requires. Thanks to Ventiva Ionic Cooling Engine (ICE), Tx has no moving parts and is noiseless, making it the first commercial application of an air-moving technology for thermal management only theorised about for decades. Developed over 10 years, ICE uses the principle of ionisation to move air silently.